



TRAINING PROGRAMME

RHCSA Rapid Track

The complete RHCSA objectives on RHEL 10, for administrators who already use Linux

48 HOURS

FOR Linux users and administrators preparing the RHCSA exam in one course

PREPARES FOR RHCSA (EX200, RHEL 10)



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Overview

This programme covers the RHCSA objectives on Red Hat Enterprise Linux 10 in one course, for participants who already work with Linux. A fast recap of the essentials leads to the system administration tasks the exam tests: networking and SSH, software, scheduling, scripting, firewalld, SELinux, partitions and LVM, NFS and autofs, permission troubleshooting and boot recovery. Rootless containers with Podman are taught as a RHEL skill outside the exam. Two timed mock EX200 exams measure readiness, and the programme closes with a final project: a full server build from a bare virtual machine against a written task list, in exam conditions.

At a glance

#	PHASE	FOCUS	HOURS
01	Core Administration	Essentials recap, system operation, networking and SSH	9 h
02	Software, Automation & Firewall	dnf and Flatpak, archives, scheduling, shell scripts, firewalld	6 h
03	Security & Storage	SELinux, partitions and LVM, NFS and autofs, ACLs	9 h
04	Boot, Containers & Mock Exams	Boot recovery, Podman, two timed EX200 mock exams	12 h
05	Final Project	A full RHCSA-style server build from a bare virtual machine	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		48 h

Key facts

AUDIENCE	Linux users and administrators preparing the RHCSA exam in one course
PREREQUISITES	Working Linux experience: command line, files, users, permissions, services, vim; recapped quickly, not taught. Beginners take RHCSA Part 1: Red Hat System Administration I and RHCSA Part 2: Red Hat System Administration II instead
DURATION	48 h: 42 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on RHEL 10; two timed mock exams
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM, hardware virtualization enabled) with two RHEL 10 virtual machines with spare virtual disks, built from the trainer's images and registered with each participant's no-cost Red Hat Developer Subscription for Individuals; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase, two timed mock EX200 exams and a final project graded against a task list; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can carry out the RHCSA objectives on RHEL 10 under timed conditions and are ready to sit the EX200 exam

PHASE 01

9 H

Core Administration

A fast recap of the command line, files and permissions brings the group to the same level, then participants operate the system: processes, tuning, services, logs and time, network configuration and key-based SSH.

Validated lab: a server with users, permissions, a persistent journal, static networking and key-only SSH, built in a set time.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Essentials Recap	Command line, files, links, users, password ageing, permissions, special permissions, documentation	Lecture + Lab	3 h
1.2	System Operation	Processes, process priorities (nice, renice), tuning profiles, services, logs and persistent journal, chrony	Lecture + Lab	3 h
1.3	Networking & SSH	nmcli, IPv4 and IPv6, name resolution, key-based SSH, secure file transfer	Lecture + Lab	3 h

PHASE 02

6 H

Software, Automation & Firewall

Participants install software from repositories and Flatpak, archive files, schedule jobs with cron, at and systemd timers, write the short shell scripts the exam asks for, and open services through firewalld. Validated lab: a script scheduled by a systemd timer that archives a directory, and a service reachable through the firewall.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Software & Archives	dnf, RPM repositories, Flatpak, tar and compression	Lecture + Lab	1.5 h
2.2	Scheduling	cron, at, systemd timer units	Lecture + Lab	1.5 h
2.3	Shell Scripting	Conditionals, loops, script arguments, processing command output	Lecture + Lab	1.5 h
2.4	Firewall	firewalld and firewall-cmd	Lecture + Lab	1.5 h

PHASE 03

9 H

Security & Storage

Participants fix SELinux denials with contexts, booleans and port labels, build storage from raw disks with partitions, LVM and swap, mount network storage on demand, and solve permission problems with ACLs. Validated lab: an extended logical volume, automounted NFS home directories and a web service on a custom port with SELinux enforcing.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	SELinux	Modes, contexts, booleans, port labels, troubleshooting	Lecture + Lab	3 h
3.2	Storage	GPT partitions, LVM, extending volumes, swap, VFAT, ext4 and XFS, mounts by UUID	Lecture + Lab	3 h
3.3	NFS & autofs	Network mounts, automounting	Lecture + Lab	1.5 h
3.4	Permission Troubleshooting	Default permissions, diagnosing permission problems, ACLs	Lecture + Lab	1.5 h

PHASE 04

12 H

Boot, Containers & Mock Exams

Participants control boot targets and the bootloader and recover a lost root password, run rootless Podman containers as systemd services, then sit two timed mock EX200 exams, each followed by a debrief of the tasks that failed. Validated lab: a mock exam passed within the time limit.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Boot & Recovery	Targets, bootloader, root password reset	Lecture + Lab	3 h
4.2	Containers with Podman	Rootless containers, systemd integration (a RHEL skill, no longer in the RHCSA 10 exam)	Lecture + Lab	3 h
4.3	Mock Exam 1	Timed EX200 mock exam on RHEL 10 and debrief of the failed tasks	Lab	3 h
4.4	Mock Exam 2	Second timed EX200 mock exam on a new task set and final debrief	Lab	3 h

PHASE 05

6 H

Final Project

Each participant builds a server from a bare RHEL 10 virtual machine against a written task list, as in the exam, then checks and documents the result before it is graded.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Task List & Build	Work through the written task list on a bare RHEL 10 virtual machine: networking, users, storage, SELinux, firewall, scheduling, NFS	Project	3 h
5.2	Verify & Document	Reboot test, self-check of every task, notes on the choices made	Project	1.5 h
5.3	Grading & Review	Grading against the task list, review of failed tasks, plan for the exam	Project	1.5 h

FINAL PROJECT

RHCSA-Style Server Build

A bare RHEL 10 virtual machine turned into a working server from a written task list covering the RHCSA objectives, with every task still correct after a reboot.

STACK RHEL 10, Bash, systemd, NetworkManager, dnf, SELinux, LVM, XFS, NFS, autofs, firewalld

DELIVERABLE Configured server, self-check notes, graded task list